The decision brief

A reproducible portfolio analysis of the supplied booking sample

WHAT THIS ANALYSIS ESTABLISHES

The sample shows concentrated booking value and measurable channel differences. Its chronology is too inconsistent for operational decisions: 59.11% of records place travel before booking.

ACCEPTED BOOKINGS

19,301

One row per unique Booking_ID

RECORDED BOOKING VALUE

INR 51.28m

All statuses; not realized revenue

CANCELLATION STATUS RATE

3.78%

729 of 19,301 accepted bookings

MEAN REPORTED OCCUPANCY

79.93%

Booking-weighted; not fleet utilization

01 Value is concentrated in two routes.

Delhi - Mumbai and Delhi - Howrah account for 66.01% of recorded booking value and 39.83% of bookings. This describes the sample mix, not profitability.

02 Website leads the observed channel mix.

IRCTC Website accounts for 53.47% of bookings and 54.03% of recorded booking value. Conversion rates cannot be calculated without traffic or funnel data.

FIRST PRIORITY: RECONCILE THE DATES

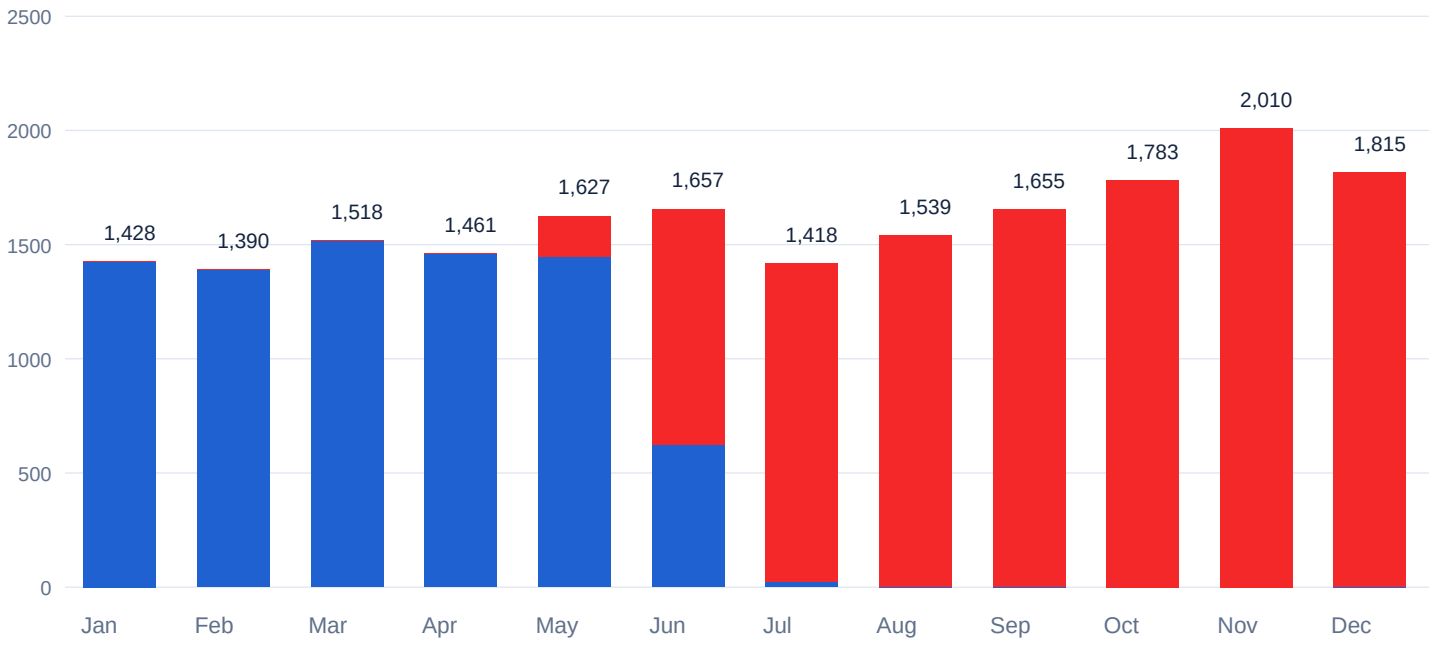
11,408 records have negative booking lead time. Preserve these rows for source reconciliation; use the 7,893-row valid-chronology cohort only as a sensitivity view.

Demand needs a time check

Monthly booking records, with impossible chronology made visible

MONTHLY BOOKINGS BY BOOKING_DATE

Blue = travel on/after booking | Red = travel before booking



November is the peak booking month in the accepted sample.

November has 2,010 records and INR 5.10m recorded booking value. December has 1,815 records, a -9.70% month-over-month change.

The valid-date cohort has a different coverage window.

Booking dates span January to December 2025, while travel dates span only 2 May to 15 July 2025. Every August-December booking therefore fails chronology. A filtered trend cannot represent the same full year.

INTERPRETATION RULE

Describe these as observed booking-date patterns. Do not infer seasonal travel demand, year-over-year growth, or a causal explanation for a monthly rise. MoM comparisons need one selected calendar month and its complete prior month.

Where booking value sits

Accepted records, all statuses; descriptive commercial mix

CHANNEL MIX

Channel	Bookings	Recorded value	Value share	Cancelled
IRCTC Website	10,320	INR 27.70m	54.03%	3.76%
IRCTC App	6,642	INR 17.54m	34.20%	3.76%
Authorised Partners	2,339	INR 6.04m	11.77%	3.89%

Cancellation percentages describe status at extraction; they do not measure the value or timing of refunds.

ROUTES RANKED BY RECORDED BOOKING VALUE

Decision hypothesis: validate the two highest-value routes first, then test whether their value concentration persists after reconciling dates and controlling for distance, class and passenger count.

Top six shown; all eight routes remain in the underlying data and analytical outputs.

Data quality is a finding

Preserve evidence, flag inconsistencies, constrain the conclusions

INTEGRITY AND MEASUREMENT CHECKS

Check	Observed result	Status
Record grain	19,301 unique accepted IDs after duplicate/conflict controls	Pass
Source controls	1 exact duplicate removed; 698 conflicting-ID records quarantined	Pass
Value arithmetic	All totals equal the four charge components	Pass
Chronology	11,408 travel dates precede booking dates	Issue
Operational grain	3,833 of 14,482 train/date groups have conflicting occupancy values	Issue
Train metadata	All 540 train numbers map to conflicting names/routes/zones	Issue

Why occupancy needs a careful label.

The row mean is 79.9318%, but a train/date is not unique and may carry multiple occupancy values. More frequently booked services get more weight. This is not a capacity-weighted utilization measure.

Why trains are counted, not inferred to have operated.

The sample contains 540 distinct train numbers. A distinct number is not a journey or a service-run count; the data lacks a validated operational run identifier.

Acceptance policy: from 20,000 source rows, remove 1 exact duplicate and quarantine all 698 rows in 349 conflicting-ID groups. Preserve every excluded row and flag retained chronology issues.

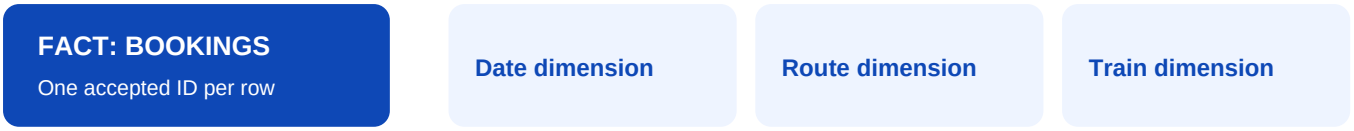
Definitions make it reproducible

Use the same cohort, date basis and grain in every output

METRIC CONTRACT

Metric	Definition
Bookings	Count unique Booking_ID. Default: 19,301 accepted records.
Recorded booking value	Sum Total_Revenue_INR over all selected statuses; not profit or cash receipts.
Cancellation status rate	Cancelled booking count / all selected booking records.
Supplied user IDs	Distinct User_ID represented in selected bookings; not registrations or monthly active app users.
Train numbers	Distinct Train_Number represented in selected bookings; not completed service runs.
Reported occupancy	Arithmetic mean of row-level Occupancy_Percentage, explicitly booking-weighted.
Lead time	Travel_Date minus Booking_Date in days; analysis restricted to nonnegative values.
Valid chronology	Travel_Date >= Booking_Date. Sensitivity cohort only; not a corrected dataset.

A BOOKING-GRAIN ANALYTICAL MODEL



Date, route and train dimensions join many bookings to one key. User IDs, channel, class, status, charges, passenger count, occupancy and delay stay on the fact. Only consistent type and capacity describe each train number. Relationship tests prevent duplicate value.

Full field definitions, model details and limitations are documented in the project files.

A practical next-step agenda

Testable hypotheses for a portfolio, with no fictional business impact

RECOMMENDED SEQUENCE

1 Reconcile chronology with the source owner.

Ask how Booking_Date and Travel_Date were generated or extracted. Obtain the true event keys, timezone and extract timestamp. Success means documenting the cause and eliminating impossible chronology with source-backed corrections.

2 Validate the high-value route concentration.

Prioritize Delhi - Mumbai and Delhi - Howrah, which contribute 66.01% of sample recorded booking value. Compare matched distance, travel class and passenger cohorts before interpreting pricing differences.

3 Investigate cancellation patterns without assuming cause.

Join status history, actual refund events and fees. Compare like-for-like routes and lead-time bands; test whether channel differences remain. A lower observed rate alone does not prove a better customer experience.

4 Build an operational capacity view only after repairing grain.

Obtain one row per service run and class with offered seats, occupied seats, service date and cancellations. Then calculate occupied seats / offered seats at a consistent grain and assess demand-capacity opportunities.

HOW TO PRESENT THIS WORK

Lead with the business question, show the 19,301-record accepted baseline, explain the chronology finding, and demonstrate a cohort/filter change. End with the evidence needed to turn descriptive patterns into a decision.

SOURCE & REPRODUCIBILITY

Source: user-supplied educational IRCTC archive. Inspiration: DataCliq tutorials, linked in docs/SOURCES.md. Metrics use the accepted cohort, reconciled to the preserved source input. See docs for definitions, a case study and interview preparation.